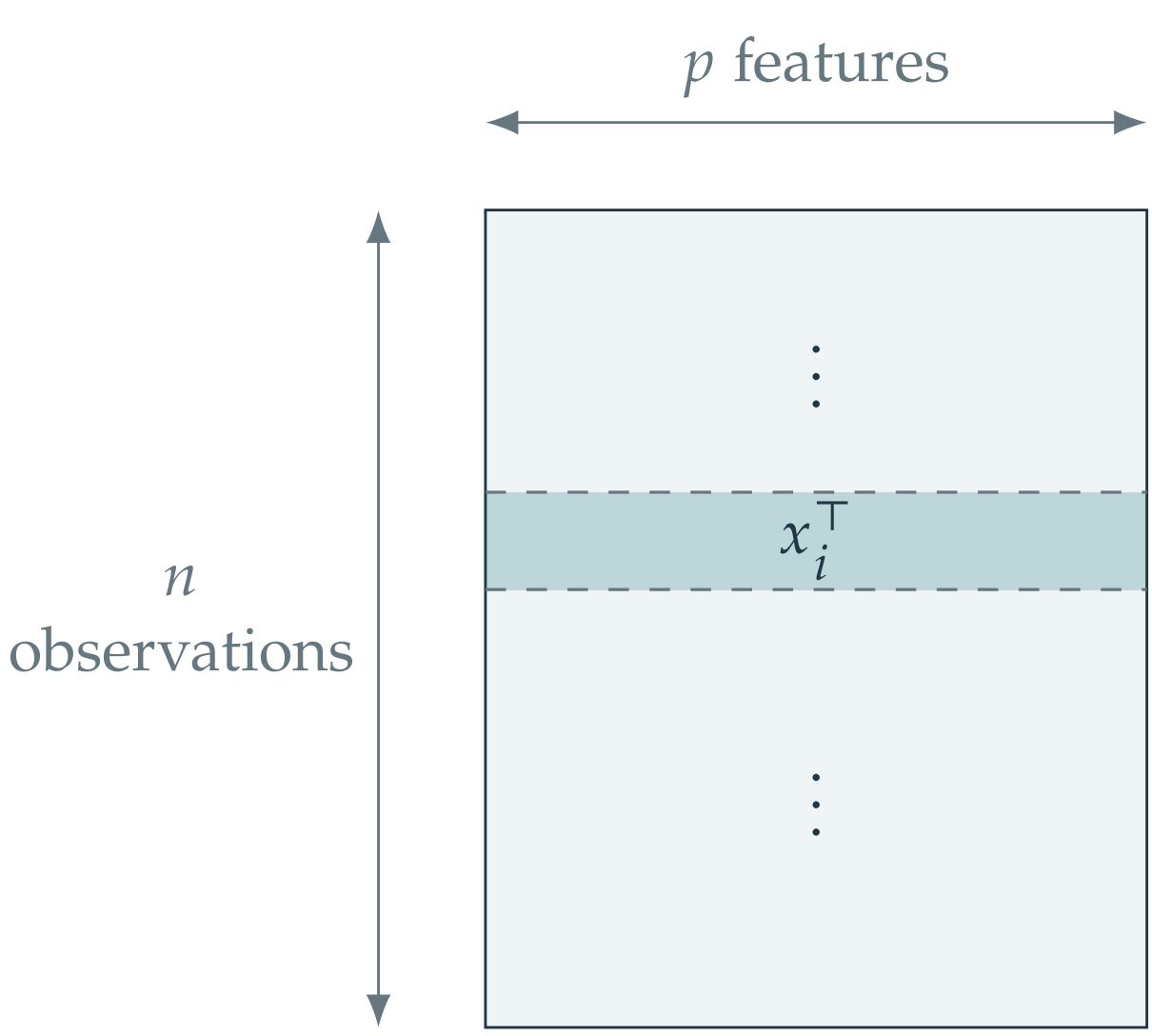

$$X = \widetilde{X} \in \mathbb{R}^{n \times p}$$

$$\widehat{m} = \frac{1}{n} \sum_{i=1}^{n} x_i = 0$$

$$\widehat{C} = \frac{1}{n} X^\top X$$

$$X = \sqrt{n}\, U \operatorname{diag}(\sigma) V^\top$$